SoPhyFit-Evap: Scientific Background and User Guide

Soil physics fitting tool for evaporation-method retention and conductivity data

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1. Introduction

SoPhyFit-Evap fits the soil water retention curve θ(h) and the unsaturated hydraulic conductivity curve K(h) to data measured with the evaporation method (HYPROP). One model describes both curves, and both data sets are fitted together. Every parameter and every derived quantity is reported with a two-sided 95% confidence interval. The program is the companion of SoPhyFit, which fits retention data from pressure plates and similar methods; both share the same model library and the same approach to uncertainty.

The evaporation method gives many retention points and, at the same time, conductivity points in the range where tensiometers work (roughly pF 2 to pF 3.5, with extended tensiometers somewhat drier). HYPROP-FIT, the software delivered with the instrument, evaluates the raw measurements and fits models to the result. SoPhyFit-Evap starts from the evaluated data exported by HYPROP-FIT (or from any workbook with a retention and a conductivity sheet) and adds:

- **Treatments and replicates:** each file is fitted, and all replicates of a treatment are fitted together (pooled), so that treatments can be compared with their uncertainty.
- **Fixed parameters per treatment:** a parameter that is not fitted (for example a measured Ks) gets one value per treatment. The default is the value HYPROP-FIT stored in the files.
- **Confidence intervals on everything:** parameters, θ(h) and K(h) bands, field capacity by four methods, wilting point, plant-available water, user-defined points and pore-class volumes.
- **Free choice of units** for display and output, and figures and tables ready for reports.

This guide explains the scientific basis of each step (Sections 3 to 7) and how to use the program (Sections 8 to 10). Section 11 describes the software and its tests, Section 12 how to cite it, Section 13 the limits of use.

2. Workflow overview

The program is a wizard. The list on the left shows the pre-processing steps; steps not reached yet are gray. The list on the right opens the result pages after a run. Two steps appear only when they apply: **Fixed Values** when at least one parameter is held fixed, and **Pore Classes** when the pore-size distribution is switched on.

1. **Select Directory.** Name the project, choose the folder with the Excel files and describe how the data are stored (sheets, columns, units). *Load Tutorial 1* opens six example files.
2. **Assign Treatments.** Give each file a treatment and a replicate id.
3. **Define Problem.** Units for display and output, fit mode, data weights, options.
4. **Model.** One of six models for both curves.
5. **Model Parameters.** Starting values, bounds, fitted or fixed.
6. **Fixed Values.** Values of the fixed parameters for each treatment.
7. **Pore Classes.** Pore-diameter classes for the pore-volume table.
8. **Specific Points.** Field-capacity method, wilting point, extra points.
9. **Run.** Fit every replicate and every treatment, write all files.

After the run: **Graphic Summary** (all figures), **Fitting Results** (parameters and statistics) and **Specific Points** (derived quantities). Every page has a Help button; Help > User Guide opens this document. The **Cite Us** tab in the menu bar gives the citation of the installed version (Section 12). Project settings are saved in `<project>_project.json` in the data folder and can be reopened with *Open Project*.

3. The data

3.1 The evaporation method

In the evaporation method a saturated soil core is sealed at the bottom and left to dry from the top. Two tensiometers at different heights record the matric head, and the core is weighed at intervals. The retention curve follows from the mean head and the mean water content of the core. The conductivity follows from the water flux between the tensiometers, computed from the mass loss, and the head gradient (Darcy-Buckingham law). Under the assumption of linear head and water-content profiles this is the simplified evaporation method of Schindler (1980), analyzed in detail by Peters and Durner (2008). Peters et al. (2015) added vapor, film and corner flow to the interpretation. HYPROP-FIT performs this evaluation (Pertassek et al., 2015).

Two properties of the data shape the fitting:

- **Conductivity is measured only in a limited range,** typically between pF 2 and pF 3.5. Near saturation the gradients are too small to compute K. The saturated conductivity Ks is therefore an extrapolation when it is fitted, and it is strongly correlated with the connectivity parameter τ. A Ks measured independently (for example with the KSAT instrument) should be used as a fixed value when available.
- **Retention points are many and dense,** conductivity points are fewer. The two data sets have different units and errors. Their weights in the fit (Section 5.1) decide how much each one shapes the parameters.

Dew-point measurements (WP4C) extend the retention curve to the dry range. HYPROP-FIT stores them in the sheet *Evaluation-WP4 manual*; SoPhyFit-Evap adds them to the retention data when the sheet holds numbers.

3.2 File formats

HYPROP-FIT export (default). The program reads

Sheet	Columns used	Unit in the file
Evaluation-Retention Θ(pF)	pF [-], Water Content [Vol%]	pF, %
Evaluation-Conductivity K(pF)	pF [-], log 10 K [cm/d]	pF, log10(cm/d)
Evaluation-WP4 manual (optional)	pF [-], Water Content [Vol%]	pF, %
Fitting-Parameter value (optional)	Parameter, Value, Unit	as stated

Any other workbook. Retention data in one sheet and conductivity data in another, each with a head column and a value column. On the *Select Directory* page the sheet names, the column headers and the units are set: head in pF, cm, hPa, kPa, MPa, bar or m; water content in % or

cm³/cm³; conductivity as log10(cm/d) or in cm/d, cm/h, cm/min, cm/s, mm/h, mm/d, m/d or m/s. A blank header means the first (head) and second (value) column. An empty conductivity sheet name means retention data only; the conductivity parameters are then not fitted.

All values are converted to internal units when the files are read: h in cm (positive suction), θ in cm³/cm³, K in cm/d. Rows with missing values are dropped, and K ≤ 0 is dropped because K is fitted on a log scale. Water contents outside 0 to 1 cm³/cm³ after conversion raise an error, which catches a wrong unit setting (for example % read as cm³/cm³).

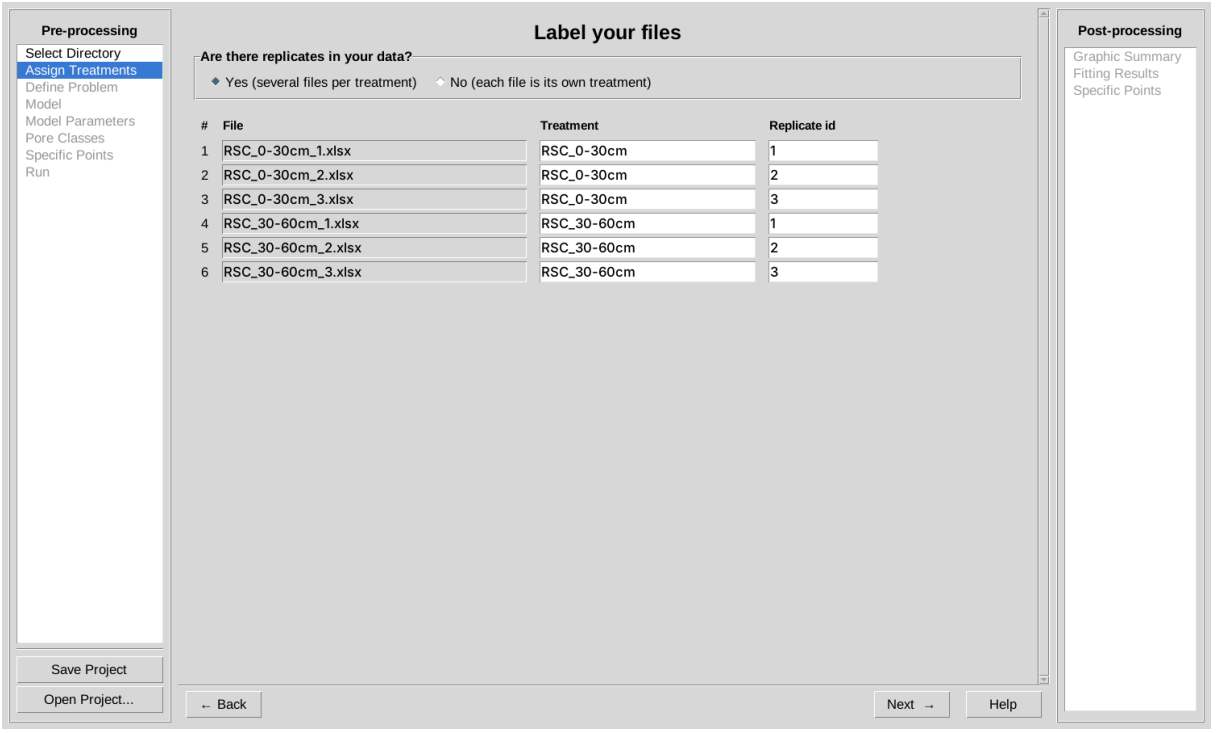


Figure 3.1. Assign Treatments. Defaults come from the file names: the text before the last underscore is the treatment, the text after it the replicate.

3.3 Units for display and output

On the *Define Problem* page the units of the output are chosen: h in cm, hPa or kPa; θ in cm³/cm³ or %; K in cm/d, cm/h, mm/h, mm/d, m/d or m/s. All values typed on later pages (bounds, fixed values, heads of specific points) are in these units, and all tables, figures and files use them. 1 cm of water corresponds to 98.0665 Pa, so 1 hPa = 1.0197 cm and 1 kPa = 10.197 cm.

Parameters are rescaled exactly: θr and θs like θ; α, α1, α2 like 1/h; hm like h; Ks like K. pF_dry stays log10 of h in cm. The head axis of the figures is pF or |h| on a log scale.

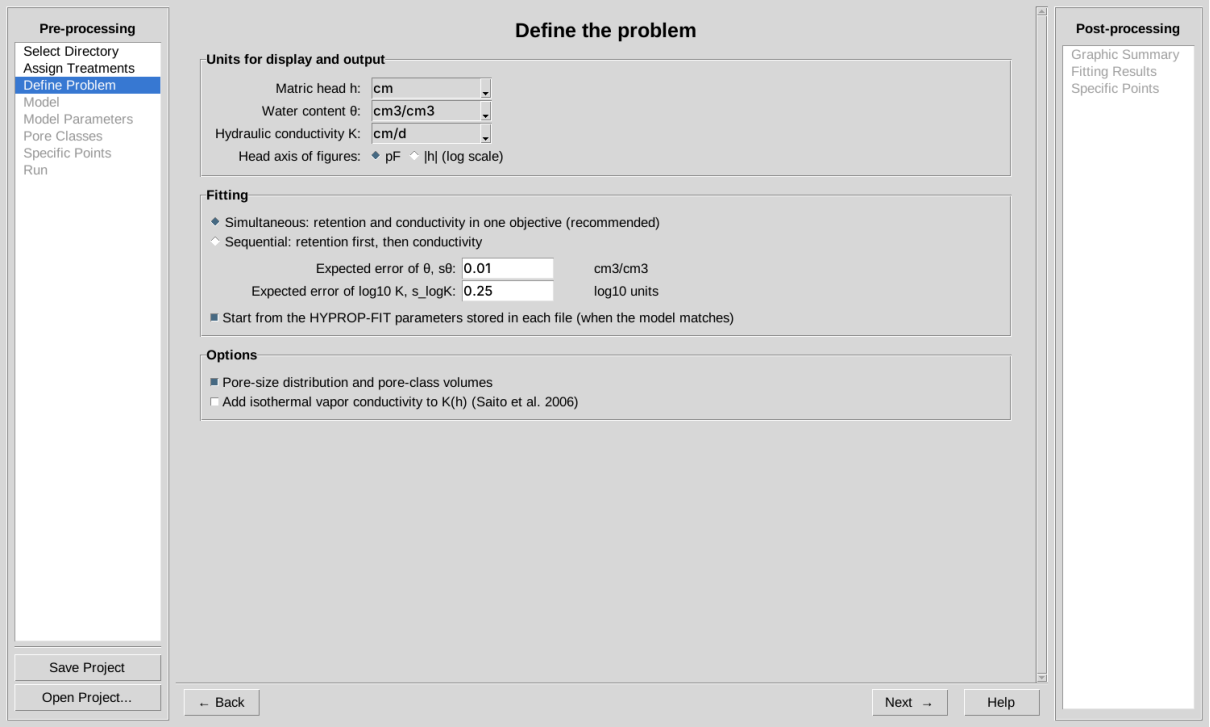


Figure 3.2. Define Problem page: units, fit mode, expected errors of θ and log10 K, starting values from HYPROP-FIT, and options.

4. Hydraulic models

Each model gives the effective saturation S(h) and, through the pore-bundle model of Mualem (1976), the relative conductivity. With θr and θs the residual and saturated water contents,

$$\theta(h) = \theta_r + (\theta_s - \theta_r)\, S(h)$$

and, for the non-PDI models,

$$K(h) = K_s\, S^{\tau} \left[\frac{\int_0^S h^{-1}\, dS}{\int_0^1 h^{-1}\, dS} \right]^2 .$$

τ (often called l) describes pore connectivity and tortuosity; Mualem proposed 0.5. With HYPROP data τ is usually fitted and acts as an empirical shape parameter; negative values are common and admissible within limits (Peters et al., 2011, derived constraints on τ that keep the functions consistent).

4.1 van Genuchten (VG)

$$S(h) = \left[1 + (\alpha h)^{n}\right]^{-m}, \quad m = 1 - 1/n$$

$$K(h) = K_s\, S^{\tau} \left[1 - \left(1 - S^{1/m}\right)^{m}\right]^2$$

(van Genuchten, 1980). The standard unimodal model. The retention curve approaches θr only asymptotically, and K falls very steeply in the dry range, so data beyond pF 4 are often poorly described.

4.2 Brooks and Corey (BC)

$$S(h) = (\alpha h)^{-\lambda} \text{ for } \alpha h > 1, \quad S = 1 \text{ otherwise}$$

with the Mualem conductivity in closed form, $K = K_s S^{\tau + 2 + 2/\lambda}$ (Brooks and Corey, 1964). 1/α is the air-entry head. The sharp air entry suits coarse, narrowly graded material.

4.3 Kosugi (KO)

$$S(h) = \tfrac{1}{2}\operatorname{erfc}\left[\frac{\ln(h/h_m)}{\sqrt{2}\,\sigma}\right]$$

$$K(h) = K_s S^{\tau}\left\{\tfrac{1}{2}\operatorname{erfc}\left[\operatorname{erfc}^{-1}(2S) + \frac{\sigma}{\sqrt{2}}\right]\right\}^2$$

(Kosugi, 1996). The retention curve of a lognormal pore-size distribution with median head hm and width σ.

4.4 Bimodal van Genuchten (Durner)

$$S(h) = w_1\left[1 + (\alpha_1 h)^{n_1}\right]^{-m_1} + w_2\left[1 + (\alpha_2 h)^{n_2}\right]^{-m_2}, \quad w_1 = 1 - w_2$$

(Durner, 1994), with the Mualem conductivity of the combined pore system in closed form (Priesack and Durner, 2006). For structured or aggregated soils whose retention curve shows two steps. It has three parameters more than VG, which the data must support.

4.5 PDI models (VG-PDI and bimodal PDI)

The PDI model system (Peters, 2013; Iden and Durner, 2014; Peters, 2014) splits the water content into a capillary and an adsorbed (non-capillary) part,

$$\theta(h) = (\theta_s - \theta_r)\, S_c(h) + \theta_r\, S_{nc}(h),$$

where S_c is the capillary saturation, rescaled so that it reaches zero at oven dryness $h_0 = 10^{pF_{dry}}$ (pF_dry = 6.8), and S_{nc} decreases log-linearly from 1 to 0 between the air-entry region and h_0. In the PDI models θr is therefore the maximum adsorbed water content, not a residual water content, and θ reaches 0 at oven dryness.

The conductivity is the sum of capillary and film (non-capillary) flow:

$$K(h) = K_s\left[(1-\omega)\, K_{r,c}(h) + \omega\, K_{r,nc}(h)\right], \qquad K_{r,nc} = \left(\frac{h_0}{h_a}\right)^{a\left(1 - S_{nc}\right)}$$

$K_{r,c}$ is the Mualem conductivity of S_c; h_a is the head where the adsorption term starts; ω is the share of film flow at saturation; a is the slope of log K versus log h in the film-flow range, −1.5 by default following Tokunaga (2009) and Peters (2013). In SoPhyFit-Evap ω is fitted and a is held fixed by default, as in HYPROP-FIT. The PDI models describe the flattening of K(h) in the dry range seen in many HYPROP data sets, and the dry end of the retention curve.

The bimodal PDI model uses the Durner capillary part with the same adsorption and film terms.

4.6 Choosing a model

Figure 4.1 shows the six models fitted to one replicate of Tutorial 1 with all parameters free (simultaneous fit, default weights). The PDI models follow the flattening of K at pF > 2.8; the plain VG and Kosugi models cannot. The table gives the fit statistics:

Model	AICc	RMSE θ [cm³/cm³]	RMSE log10 K
van Genuchten	−180	0.0044	0.130
Brooks-Corey	−74	0.0075	0.114
Kosugi	−84	0.0059	0.238
Bimodal van Genuchten	−357	0.0019	0.082
VG-PDI	−277	0.0033	0.053
Bimodal PDI	−402	0.0019	0.043

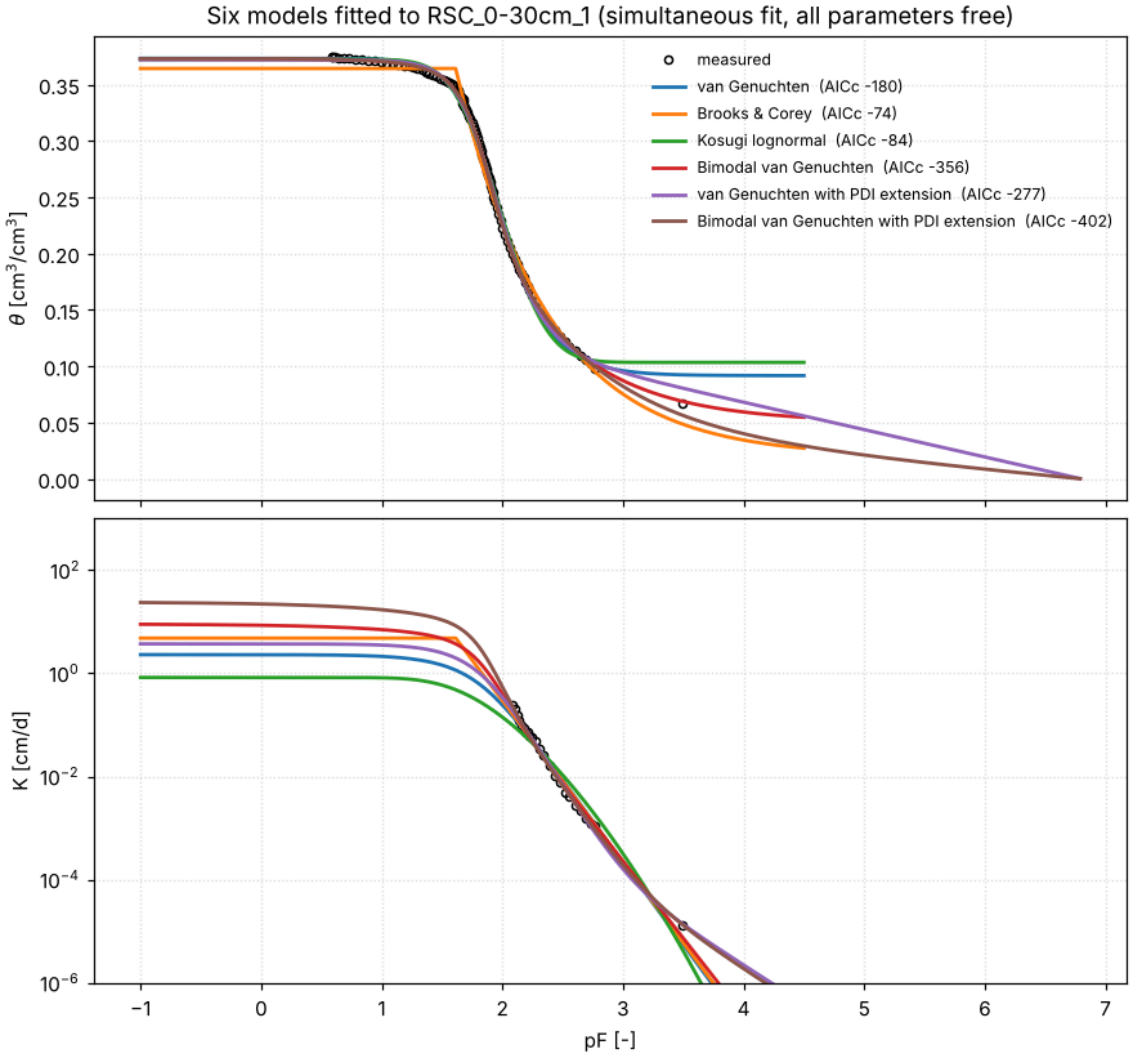


Figure 4.1. Six models fitted to RSC_0-30cm_1. AICc in the legend (lower is better).

A practical order: start with VG-PDI; use a bimodal model only when the retention data show two steps and the AICc drops clearly (by more than about 10); check that the bimodal parameters are not at bounds and not strongly correlated. A lower AICc for a bimodal model

may reflect structure, but also a slight misfit of a unimodal shape; the parameter correlations and the physical plausibility decide.

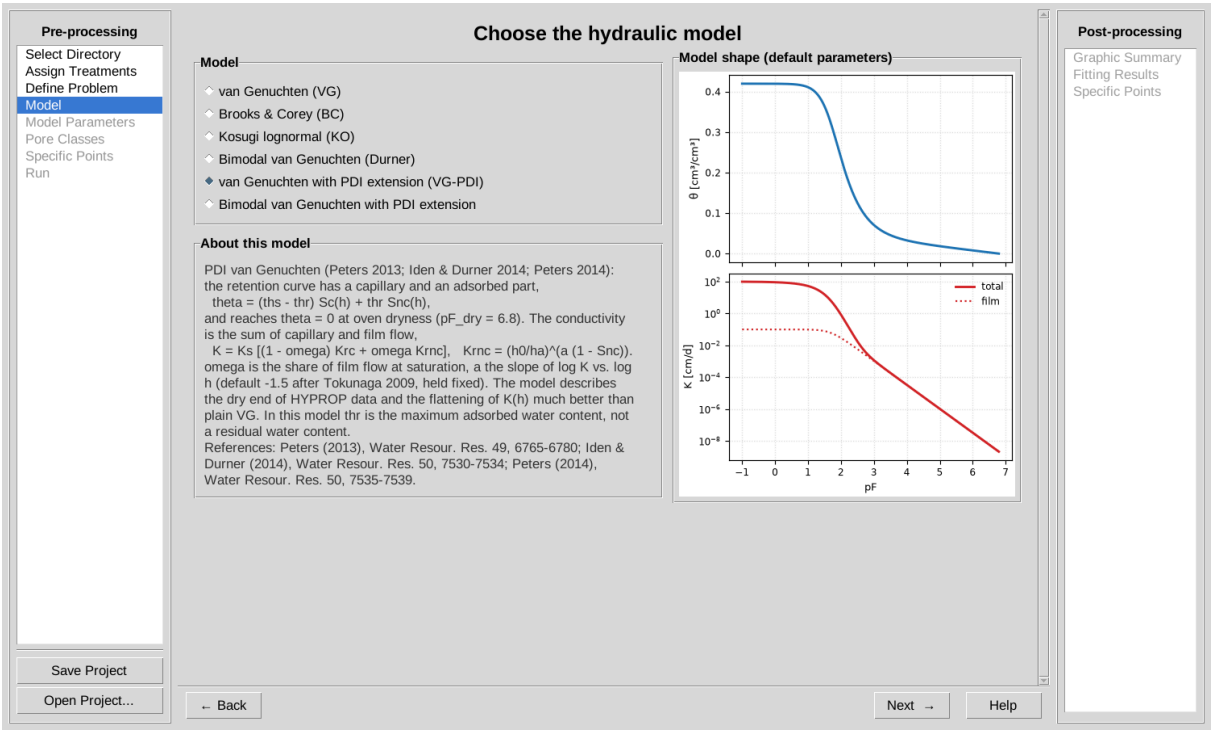


Figure 4.2. Model page with the description of the model and its shape for default parameters.

5. Fitting

5.1 Objective function and weights

The parameters **p** minimize

$$\Phi(\mathbf{p}) = \sum_{i=1}^{N_\theta} \left(\frac{\theta_i - \theta(h_i;\mathbf{p})}{s_\theta} \right)^2 + \sum_{j=1}^{N_K} \left(\frac{\log_{10} K_j - \log_{10} K(h_j;\mathbf{p})}{s_{\log K}} \right)^2 .$$

The weights are error based: s_θ and $s_{\log K}$ are the expected errors of a single measurement (Peters and Durner, 2008). Defaults are s_θ = 0.01 cm³/cm³ and $s_{\log K}$ = 0.25 (a factor of 1.8 in K). Their ratio sets the balance between the two data sets. It does not depend on how many points each set has: a large set dominates because it has many points, which is what least squares should do when all points are equally reliable. Conductivity is fitted on a log scale because it spans many orders of magnitude and its errors are roughly proportional to K.

If the actual scatter matches the expected errors, the reduced chi-square $\Phi/(N-p)$ is near 1. Values well below 1 mean the errors were set too large (the data are smoother than assumed; HYPROP retention data are often very smooth); values well above 1 mean they were set too small or the model does not fit. The confidence intervals are scaled by the reduced chi-square (Section 6), so they do not depend on the absolute size of s_θ and $s_{\log K}$, only on their ratio.

5.2 Simultaneous and sequential fit

- **Simultaneous (default).** All parameters are fitted to both data sets at once. The shape parameters (α, n, ...) then reflect both curves. This is the HYPROP-FIT approach.

- **Sequential.** First the retention parameters are fitted to θ(h); then Ks, τ, ω and a are fitted to K(h) with the retention parameters held. The retention curve is then not affected by the conductivity data. The confidence intervals are computed from the Jacobian of the full objective at the final estimate in both modes.

5.3 Optimization

The optimizer combines a global and a local step: Nelder-Mead from the starting values, then a trust-region least-squares polish (SciPy `least_squares`). Parameters that span decades (α, α1, α2, Ks, hm) are searched on a log10 scale, which makes the search independent of their magnitude. Bounds are respected in both steps. When *Start from the HYPROP-FIT parameters* is ticked and HYPROP-FIT used the same model family, the values stored in each file are the starting values for that file.

5.4 Replicates and pooled fit

Every file (replicate) is fitted on its own. For a treatment with several replicates, one more fit is made to all their points pooled; it starts from the median of the replicate estimates. This **final fit** represents the treatment in the comparison figures and tables. The spread of the replicate fits shows the variability between samples; the confidence interval of the pooled fit shows how precisely the mean curve is known. Both can be shown as error bars in the bar charts.

5.5 Fixed parameters per treatment

Unticking *Fit* on the parameter page holds the parameter fixed. The *Fixed Values* page then shows one value per treatment and fixed parameter, used for all replicates of the treatment and for its pooled fit.

The default is taken from the files:

1. If the files of the treatment contain the sheet *Fitting-Parameter value*, the mean of their values is used (geometric mean for Ks, α and hm). Parameters that keep their meaning in every model (θs, Ks, τ) are always taken; shape parameters (α, n, θr, …) only when HYPROP-FIT used the same model family.
2. Otherwise, for example with your own two-sheet files, the starting value of the parameter page is used.

The source of each default is shown under the field (*HYPROP file* or *parameter page*). Typed values override the default; *Restore defaults* undoes all edits. Values are checked against their physical range (for example Ks > 0, 0 ≤ ω ≤ 1, a < 0, 5 ≤ pF_dry ≤ 8).

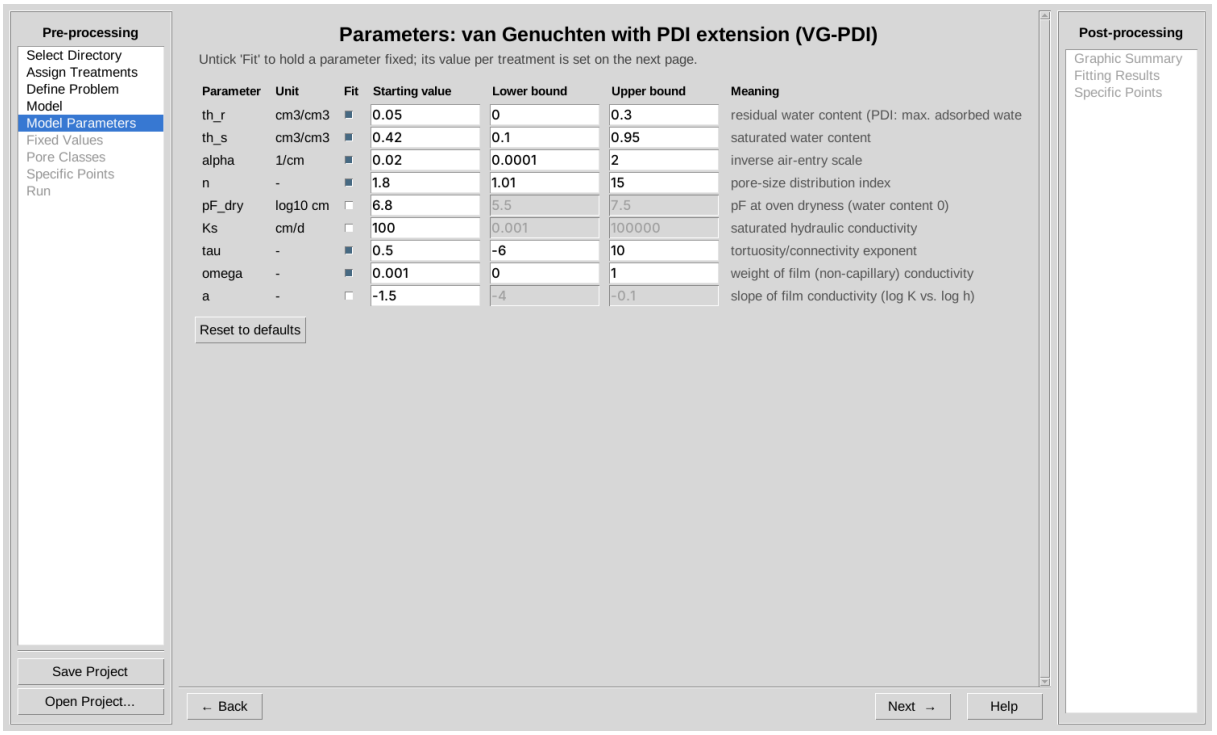


Figure 5.1. Model Parameters page (VG-PDI). Ks, pF_dry and a are fixed.

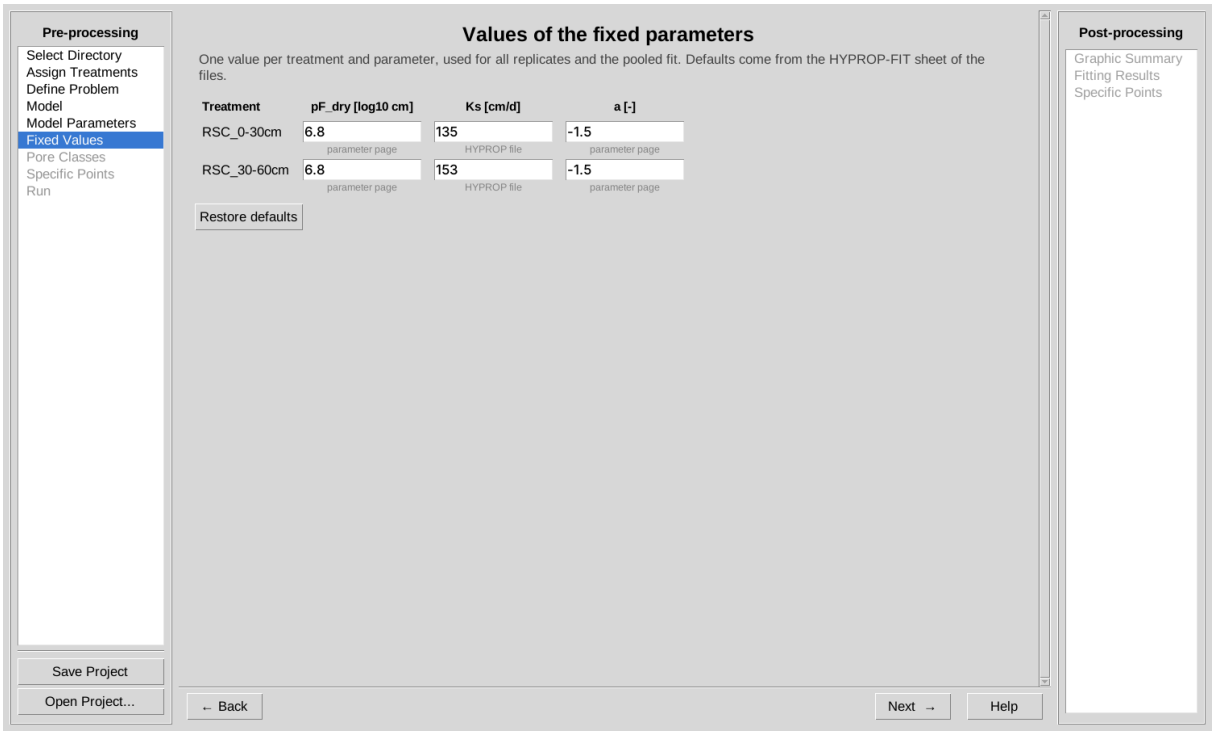


Figure 5.2. Fixed Values page. Ks per treatment defaults to the HYPROP-FIT value of the files (135 and 153 cm/d).

6. Uncertainty

6.1 Parameter covariance and confidence intervals

At the optimum $\hat{\mathbf{p}}$ the covariance of the fitted parameters is

$$\mathbf{C} = s^2 \left(\mathbf{J}^{\mathsf{T}} \mathbf{J} \right)^{-1}, \qquad s^2 = \frac{\Phi \left(\widehat{\mathbf{p}} \right)}{N - p},$$

with $\mathbf{J}$ the Jacobian of the weighted residuals (central differences), N the number of retention and conductivity points and p the number of fitted parameters. The standard error of a parameter is the square root of its diagonal element, and its 95% interval is

$$\widehat{p}_k \pm t_{0.975,\, N-p}\, \mathrm{SE} \left(\widehat{p}_k \right).$$

For any derived quantity g(**p**) (a point of the curve, field capacity, PAWC, a pore-class volume) the delta method gives

$$\mathrm{Var} \left(g \right) \approx \nabla g^{\mathsf{T}}\, \mathbf{C}\, \nabla g, \qquad g \pm t_{0.975,\, N-p} \sqrt{\mathrm{Var} \left(g \right)}.$$

Heads and conductivities are treated on a log scale, so their intervals are symmetric in log10 and always positive. PAWC = θ(FC) − θ(PWP) uses the full covariance, so the correlation of both water contents is accounted for.

These are intervals of the fitted (mean) curve for one data set, not prediction intervals for new measurements, and they do not include uncertainty of fixed parameters or of the data evaluation in HYPROP-FIT.

6.2 Checks and warnings

- **Singular covariance.** If the data cannot identify all fitted parameters (for example a bimodal model on unimodal data), $\mathbf{J}^{\mathsf{T}}\mathbf{J}$ is singular. No standard errors or intervals are reported, and the statistics carry a warning. Fix parameters or use a simpler model.
- **At bound.** A parameter within 0.1% of a bound is flagged; its interval is then not reliable.
- **Strong correlations.** Pairs with |r| > 0.9 are listed on the results page, |r| > 0.95 also in the run log. Ks and τ are typically correlated when Ks is fitted.
- **Coverage.** A Monte Carlo test (`coverage_check.py`, 300 synthetic data sets with the assumed errors) gave coverage between 0.94 and 0.98 for all parameters, θ at given heads, K at a given head, FC, PWP and PAWC, close to the nominal 0.95.

6.3 Goodness of fit and model comparison

The results report, for each fit: RMSE and R^2 of θ and of log10 K, the numbers of points, the weighted sum of squares Φ, the reduced chi-square, and

$$\mathrm{AIC} = N \ln \left(\Phi / N \right) + 2p, \quad \mathrm{AICc} = \mathrm{AIC} + \frac{2p \left(p+1 \right)}{N - p - 1}, \quad \mathrm{BIC} = N \ln \left(\Phi / N \right) + p \ln N.$$

AIC, AICc and BIC compare models fitted to the same data with the same weights; the lowest value is preferred, and differences below about 2 are not meaningful.

7. Derived quantities

7.1 Field capacity

One of four methods is chosen on the *Specific Points* page.

Method	Definition	Reference
Fixed matric head	θ at a chosen head, e.g. 330 cm (33 kPa), or 100 cm (10 kPa) for coarse soils	traditional
Inflection point	head where θ decreases fastest with log h; for VG $h = \alpha^{-1}\, m^{-1/n}$	Ferreira and Marcos (1983); Dexter and Bird (2001)
Drainage flux	head where K(h) equals a negligible flux q (unit gradient); default q = 0.01 cm/d (0.1 mm/d)	Twarakavi et al. (2009)
Assouline and Or	$h_{FC} = \alpha^{-1} [\, (n-1)/n \,]^{(1-2n)/n}$; VG and VG-PDI only	Assouline and Or (2014)

Dexter and Bird (2001) used the inflection point for the optimum water content for tillage; using it as field capacity goes back to Ferreira and Marcos (1983). The inflection point is not defined for Brooks-Corey (the curve has a kink). The flux criterion uses the fitted K(h), so it reflects the measured conductivity; Twarakavi et al. (2009) tested thresholds of 0.001 to 0.1 cm/d and recommended 0.01 cm/d. For the last three methods the field-capacity head itself depends on the parameters, and the program reports its 95% interval too.

7.2 Wilting point and plant-available water

The permanent wilting point (PWP) is the water content at a chosen head, 15849 cm (pF 4.2) by default (1500 kPa would be 15296 cm). The plant-available water content is PAWC = θ(FC) − θ(PWP). If the retention data end well before pF 4.2 (HYPROP alone often ends near pF 3.5 to 4), θ(PWP) is an extrapolation of the model; WP4 points remove this problem.

7.3 User points

θ and K at given heads, and h at given water contents, each with its interval.

7.4 Pore-size distribution and pore classes

With the Young-Laplace equation (zero contact angle) a head h drains pores of equivalent diameter $d = 4\sigma / \left(\rho_w g h \right)$, about 2970/h µm with h in cm at 20 °C. The pore-size distribution is shown as $d\theta / d \log_{10} r$ of the capillary part; its area is the drainable pore volume.

Pore classes give the water held in pores between two diameters: $V = \theta \left(h \left(d_{high} \right) \right) - \theta \left(h \left(d_{low} \right) \right)$. The default classes follow the German soil mapping guide KA5 (Ad-hoc-AG Boden, 2005): wide coarse pores > 50 µm (pF < 1.8), narrow coarse pores 10 to 50 µm (pF 1.8 to 2.5), medium pores 0.2 to 10 µm (pF 2.5 to 4.2), fine pores < 0.2 µm (pF > 4.2). User-defined classes are possible. Classes covering 0 to ∞ add up to θs.

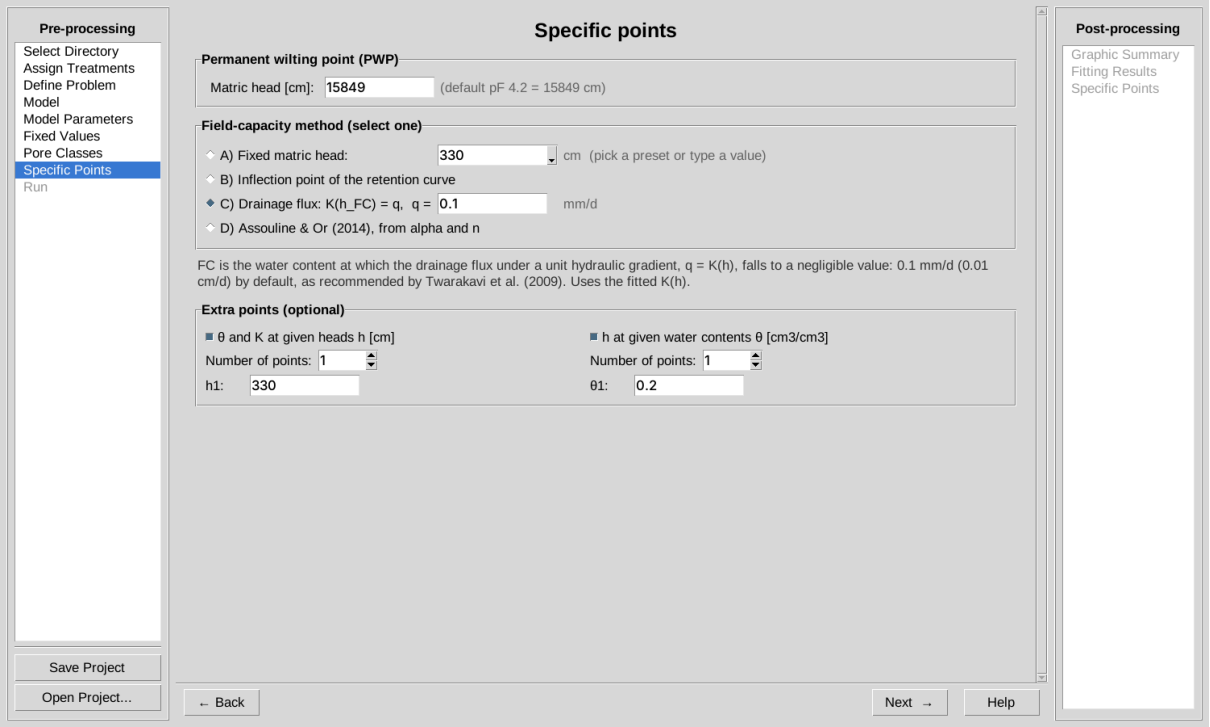


Figure 7.1. Specific Points page with the drainage-flux method and two extra points.

8. Using the program step by step

Select Directory. Type a project name and choose the data folder. The data-layout blocks describe the retention, conductivity and WP4 sheets; *Restore HYPROP-FIT defaults* resets them. *Load Tutorial 1* copies the example files to `Documents\SoPhyFit-Evap\Tutorial 1` and opens them.

Assign Treatments. Check treatment names and replicate ids. With *No replication* every file is its own treatment.

Define Problem. Units, head axis (pF or |h|), fit mode, s_θ and $s_{\log K}$ (in the chosen θ unit and in log10 units), starting values from HYPROP-FIT, pore-size distribution, and vapor conductivity (Saito et al., 2006; it matters only in the very dry range).

Model. Choose the model; the panel shows its equations, references and shape.

Model Parameters. Starting value, lower and upper bound, and *Fit* for each parameter, in the chosen units. Good practice: fix Ks when a measured value exists; keep pF_dry and a fixed in PDI models; fix θr in bimodal models unless dry data support it.

Fixed Values. Check or edit the value per treatment (Section 5.5).

Pore Classes. KA5 classes or your own; the matching head range is shown in the chosen unit.

Specific Points. PWP head, field-capacity method, extra heads and water contents.

Run. Press *Run*. The progress window lists each file read (with the number of retention, WP4 and conductivity points), each fit with R^2 and RMSE of θ and log10 K and the fitted parameters, warnings, and the files written. A file that fails is reported and skipped. The same messages go

to `sophyfit_hyprop_log.txt`. A run of Tutorial 1 (six files, two pooled fits, VG-PDI) takes about five seconds.

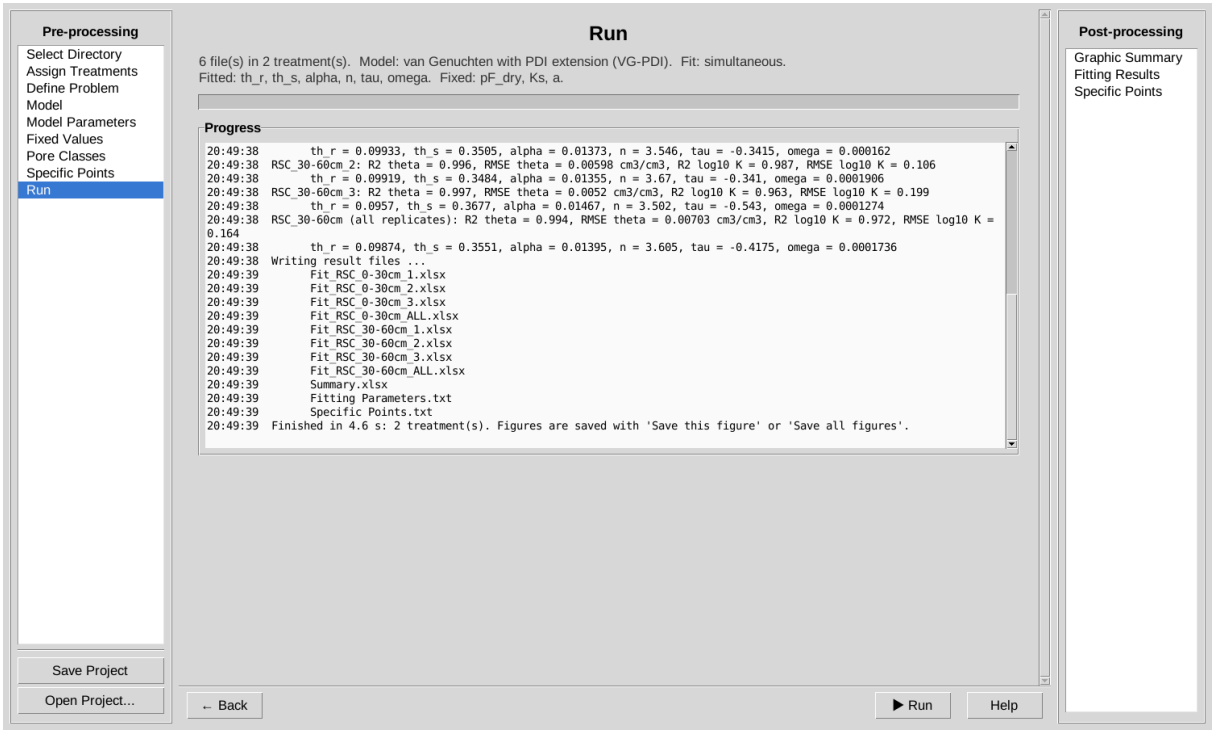


Figure 8.1. Run page after a run of Tutorial 1.

9. Results

9.1 Graphic Summary

Figures, chosen in the drop-down list or with *Previous* and *Next*:

- all treatments with their final fits;
- each treatment: replicates in colors and the fit to all replicates;
- each replicate with its own fit;
- conductivity components (capillary, film, vapor) for PDI models or with vapor on;
- pore-size distribution and pore-class volumes;
- FC, PWP and PAWC per treatment;
- θ and K at the given heads, h at the given water contents.

Retention (top) and conductivity (bottom) share the head axis. *Show 95% CI band* shades the confidence band of θ(h) and K(h). *Figure properties* sets font and line sizes, grid, save DPI, the error bars of the bar charts (SD of replicates, 95% CI of the final fit, or none), and legend, color, symbol and line style per treatment. *Labels & limits* changes axis titles and limits of the current figure only. Figures are saved only on request (*Save this figure*, *Save all figures*), as PNG in the results folder.

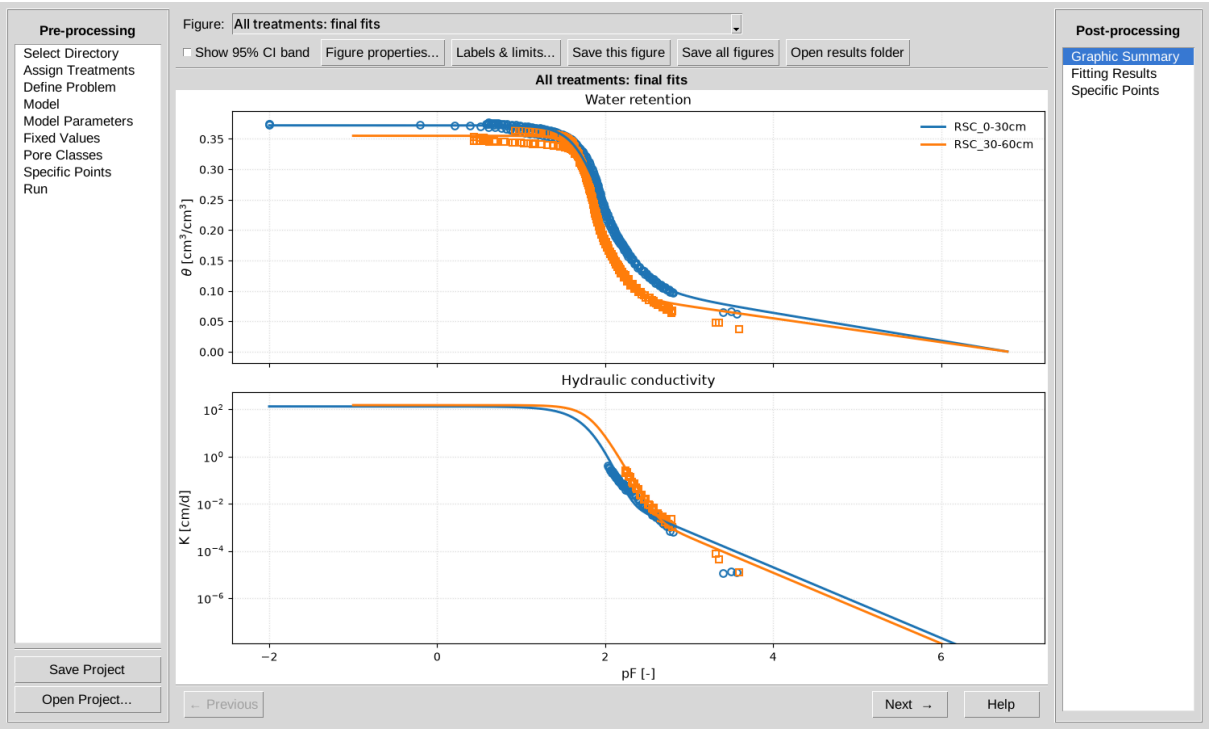


Figure 9.1. Graphic Summary: both treatments of Tutorial 1 with their final fits.

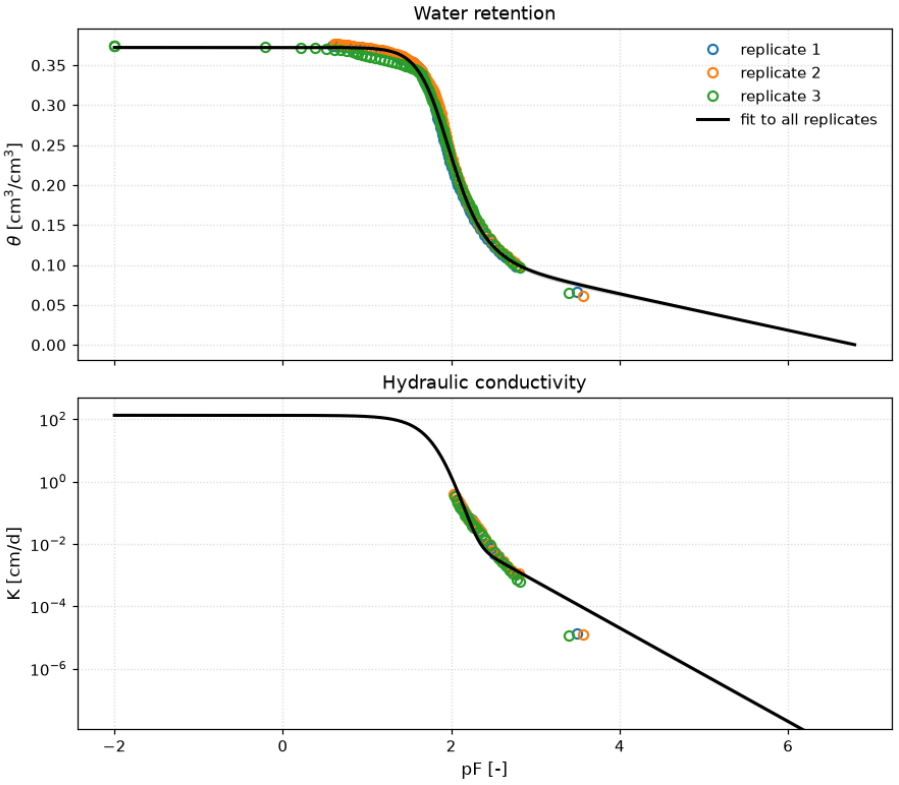


Figure 9.2. Treatment RSC_0-30cm: three replicates and the fit to all replicates (VG-PDI, Ks fixed at the HYPROP-FIT value).

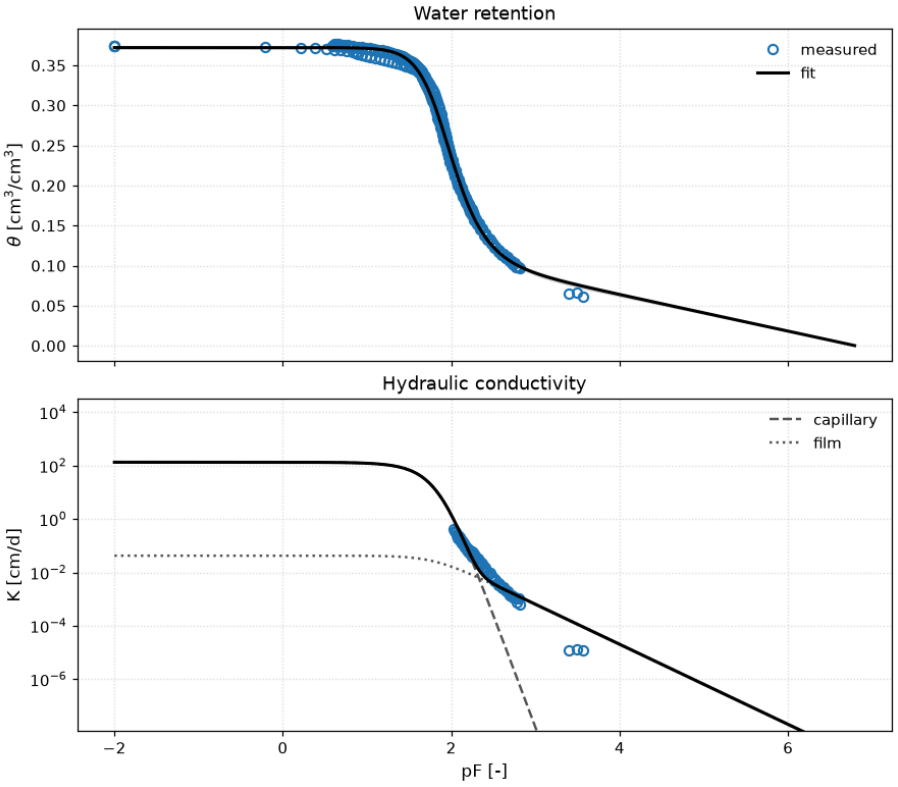


Figure 9.3. Conductivity components of the VG-PDI fit: capillary flow dominates near saturation, film flow beyond pF 2.7.

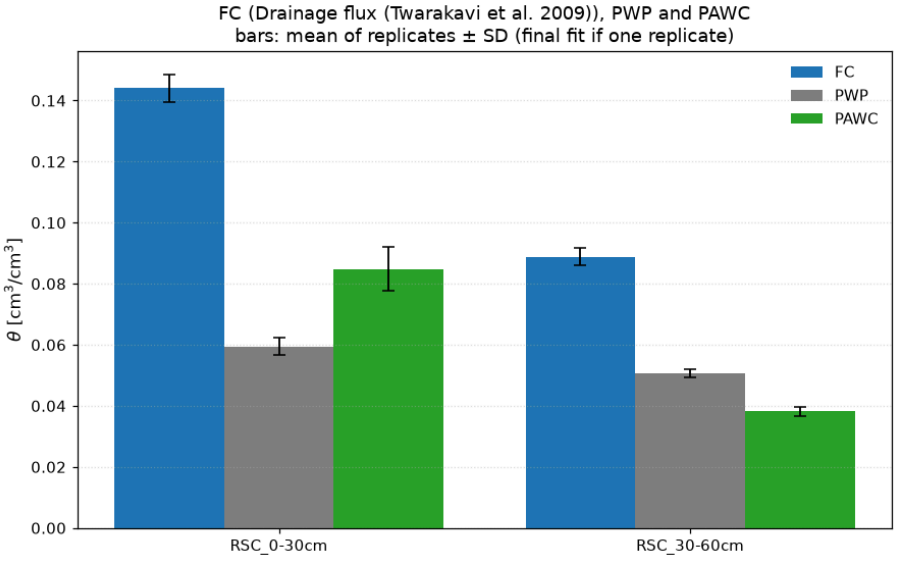


Figure 9.4. FC (drainage flux), PWP and PAWC per treatment; bars show the mean of the replicates ± 1 SD.

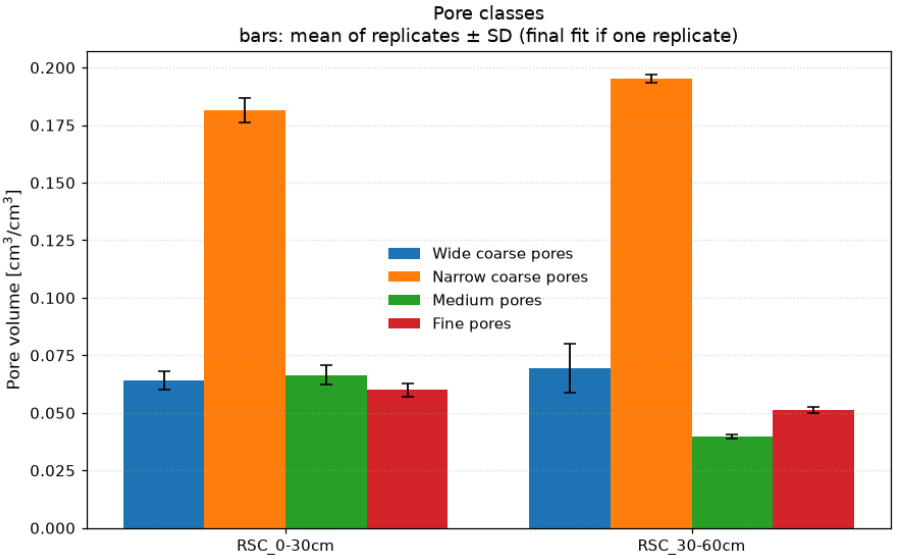


Figure 9.5. KA5 pore-class volumes per treatment.

9.2 Fitting Results and Specific Points

Fitting Results lists, for the final fits or for all fits, every parameter with unit, value, standard error, 95% interval and notes (*fixed*, *AT BOUND*), followed by the goodness-of-fit statistics and strong correlations. *Specific Points* lists FC (head and θ), PWP, PAWC, user points and pore classes with their intervals.

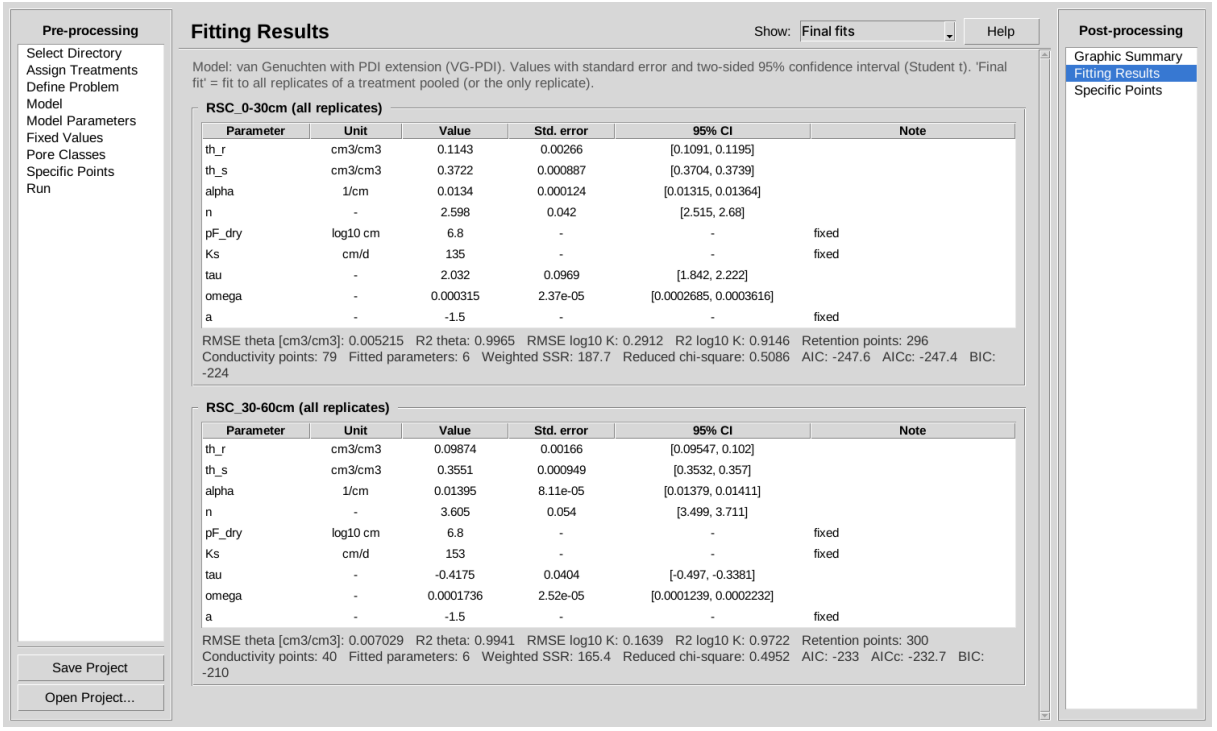


Figure 9.6. Fitting Results for the pooled fits of Tutorial 1.

9.3 Files

Results go to `<project>_Results` in the data folder (or to `Documents\SoPhyFit-Evap` if the data folder is read-only), all in the chosen units:

File	Content
Fit_*.xlsx, Fit*_ALL.xlsx	measured retention (with source HYPROP or WP4) and conductivity; fitted curves on a pF grid with 95% bands and K components; parameters with SE and CI; statistics; correlations; specific points; pore classes
Summary.xlsx	all fits side by side: parameters with CI, statistics, specific points, pore classes
Fitting Parameters.txt	report of every fit
Specific Points.txt	report of the derived quantities
sophyfit_hyprop_log.txt	run log
Figure_*.png	figures, when saved

Files with these names are replaced at each run; other files in the folder are never deleted.

10. Tutorial 1

Tutorial 1 holds six HYPROP-FIT exports of a soil sampled at two depths (RSC_0-30cm and RSC_30-60cm), three replicates each. The files contain HYPROP-FIT parameters of a van Genuchten fit with Ks fixed (135 and 153 cm/d).

Suggested settings: VG-PDI; Ks unticked (fixed), so its value per treatment comes from the files; drainage-flux field capacity. Results of the pooled fits:

	RSC_0-30cm	RSC_30-60cm
θs [cm³/cm³]	0.372 [0.370, 0.374]	0.355 [0.353, 0.357]
α [1/cm]	0.0134 [0.0132, 0.0136]	0.0140 [0.0138, 0.0141]
n [-]	2.60 [2.52, 2.68]	3.61 [3.50, 3.71]
τ [-]	2.03 [1.84, 2.22]	−0.42 [−0.50, −0.34]
R^2 θ / R^2 log10 K	0.997 / 0.915	0.994 / 0.972
FC head [cm]	223 [218, 230]	322 [312, 333]
PAWC [cm³/cm³]	0.085 [0.082, 0.088]	0.038 [0.037, 0.039]

The deeper layer has a narrower pore-size distribution (larger n), drains at a similar head, and holds less plant-available water.

11. Software design and tests

Module	Task
`HydraulicModels.py`	model equations (shared with SoPhyFit)
`water_properties.py`	water properties, Young-Laplace relation
`units.py`	units and parameter scaling
`evap_core.py`	reading, fitting, covariance, intervals, derived quantities (no GUI)
`evap_io.py`	conversion to the user's units, Excel files, reports
`SoPhyFit_Evap.py`	user interface
`updater.py`	update check

Tests: `benchmark_models.py` compares θ(h) of every model with the published reference formulas of the PDI model system (maximum deviation $2.5 \cdot 10^{-16}$); `test_hydraulic_models.py` checks the closed-form conductivities against numerical integration; `test_evap_core.py` checks reading of HYPROP and two-sheet files, WP4 points, the defaults of fixed values, the PDI conductivity mapping (K(0) = Ks, film part = ωKs), recovery of known parameters from synthetic data in both fit modes, the field-capacity methods against their analytical forms, and the pore classes; `coverage_check.py` checks the coverage of the intervals (Section 6.2).

At every start the program asks GitHub for the latest release (no data are sent). *Help > Check for Updates* checks on demand.

12. How to cite

The **Cite Us** tab shows the citation of the installed version as text and as BibTeX, each with a Copy button, links to the DOIs, and the original references of the models and of the evaporation method. Each release is archived on Zenodo with its own DOI. The program reads the DOI of the installed version from its release notes on GitHub when the tab opens (online only) and remembers it for offline use. The DOI of all versions always leads to the latest version; for a publication, cite the version you used:

> Zarebanadkouki, M. (2026). SoPhyFit-Evap: Soil physics fitting tool for evaporation-method retention and conductivity data (Version 1.0.0) [Computer software]. Technical University of Munich. Zenodo. https://doi.org/(DOI of the version)

Please also cite the model you used (Section 4) and, for the method, Peters and Durner (2008).

13. Limits of use

- The program fits evaluated data. The evaluation of the raw HYPROP measurements (linearization, flux computation, exclusion of points near the end) is done by HYPROP-FIT.
- Ks fitted from HYPROP conductivity data is an extrapolation; use a measured Ks when available.
- θ(PWP), PAWC and dry-range pore classes are extrapolations if no data reach pF 4.2.
- The intervals assume independent errors and a correct model. HYPROP retention points are closely spaced and autocorrelated, so the intervals of a single file can be optimistic;

the spread between replicates is the more realistic measure of uncertainty between samples.
- Results differ from HYPROP-FIT when weights, fixed parameters or model options differ.
- Hysteresis, swelling and shrinkage are not considered.

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